

Beta oscillations reflect supramodal information during perceptual judgment

Saskia Haegens^{a,b,1}, José Vergara^c, Román Rossi-Pool^c, Luis Lemus^c, and Ranulfo Romo^{c,d,1}

^aDepartment of Neurosurgery, Columbia University Medical Center, New York, NY 10032; ^bDonders Institute for Brain, Cognition and Behaviour, Radboud University Nijmegen, 6500HB Nijmegen, The Netherlands; ^cInstituto de Fisiología Celular-Neurociencias, Universidad Nacional Autónoma de México, 04510 Mexico D.F., Mexico; and ^dEl Colegio Nacional, 06020 Mexico D.F., Mexico

Contributed by Ranulfo Romo, November 15, 2017 (sent for review August 18, 2017; reviewed by Felix Blankenburg and Tobias H. Donner)

Previous work on perceptual decision making in the sensorimotor system has shown population dynamics in the beta band, corresponding to the encoding of stimulus properties and the final decision outcome. Here, we asked how oscillatory dynamics in the medial premotor cortex (MPC) contribute to supramodal perceptual decision making. We recorded local field potentials (LFPs) and spikes in two monkeys trained to perform a tactile-acoustic frequency discrimination task, including both unimodal and crossmodal conditions. We studied the role of oscillatory activity as a function of stimulus properties (frequency and sensory modality), as well as decision outcome. We found that beta-band power correlated with relevant stimulus properties: there was a significant modulation by stimulus frequency during the working-memory (WM) retention interval, as well as modulation by stimulus modality—the latter was observed only in the case of a purely unimodal task, where modality information was relevant to prepare for the upcoming second stimulus. Furthermore, we found a significant modulation of beta power during the comparison and decision period, which was predictive of decision outcome. Finally, beta-band spike-field coherence (SFC) matched these LFP observations. In conclusion, we demonstrate that beta power in MPC is reflective of stimulus features in a supramodal, context-dependent manner, and additionally reflects the decision outcome. We propose that these beta modulations are a signature of the recruitment of functional neuronal ensembles, which encode task-relevant information.

beta oscillations | perceptual decision making | working memory | supramodal | LFP

Perceptual decision making has been studied extensively in the monkey sensorimotor system using a vibrotactile discrimination task (1–3). Spike activity in somatosensory, (pre)motor, and prefrontal regions has been linked to various task aspects, including stimulus encoding (4, 5), working-memory (WM) maintenance (6, 7), and comparison (8–10). Population dynamics in this same paradigm have been studied using local field potential (LFP) and magnetoencephalography and electroencephalography recordings, showing oscillatory modulations in alpha (8–14 Hz) and beta bands (15–30 Hz) related to various task aspects: alpha sets the state of the system, with decreased alpha in task-relevant areas related to increased spike activity and improved performance (11, 12), while beta modulations correspond to the encoding of stimulus properties (13, 14) and the final decision outcome (refs. 15 and 16; for review, see ref. 17). Combined, these studies paint a picture of a dynamic, distributed network of areas contributing to the various stages of the perceptual decision-making process.

Recently, this framework has been extended to the multimodal case: if the subject has to discriminate stimuli from different modalities, where and how is the modality-specific information transformed into a supramodal signal, allowing for comparison across modalities? In a tactile-acoustic version of the discrimination task, it was shown that while early sensory regions only encode information from their principal modality (18), neurons in premotor cortex encode information in a supramodal sense, that is, using the same representation to maintain information from either modality in WM (19). Furthermore, an EEG study in human

subjects, using visual, tactile, and auditory unimodal flutter discrimination, reported parametric modulation of prefrontal beta power by stimulus frequency, regardless of modality, during WM retention (20). Thus, these studies suggest that task-relevant stimulus properties are encoded in a modality-nonspecific way in prefrontal regions, allowing for subsequent supramodal perceptual judgment.

Here, we asked how oscillatory dynamics contribute to the supramodal perceptual decision process. We recorded LFPs and spikes in medial premotor cortex (MPC) in two animals performing a tactile-acoustic flutter discrimination task. We studied the role of oscillatory activity—specifically, alpha and beta bands—as a function of stimulus properties (frequency and sensory modality) and decision outcome. We demonstrate that beta power in MPC is reflective of stimulus features in a context-dependent manner, and the subsequent decision outcome. This information is coded in a supramodal manner: modality information is only retained when relevant for the task at hand.

Results

We recorded LFPs from MPC (Fig. S1) in two monkeys performing a tactile-acoustic discrimination task (Fig. 1 *A* and *B* and *SI Materials and Methods*). Reaction times did not differ significantly across modality conditions, while accuracy was higher on TT than on AA/AT trials, and no significant difference was found for TT vs. TA (with T for tactile stimulus, and A for acoustic stimulus; see Fig. 1 *C–F* for further details). There was a slight bias for m28 to answer $f_2 > f_1$, while m22 had a bias to answer $f_2 < f_1$ ($P < 0.05$, Wilcoxon signed-rank test comparing performance on trials with same f_1 sorted by $f_2 - f_1$ difference; Fig. S2). Here, we studied the role of oscillatory activity as a

Significance

When a perceptual judgment has to be made comparing inputs from different sensory modalities (here, tactile and auditory), the relevant information needs to be coded in a supramodal way. We studied the role of beta oscillations in the premotor system, which had previously been linked to tactile working memory and decision making. We report that beta-band power reflects the to-be-remembered stimulus properties, and the subsequent decision outcome. This information is coded in a supramodal, context-dependent manner—modality information is retained only when relevant for the task at hand. We interpret these results in light of a recently proposed framework in which beta-band synchronization reflects the dynamic recruiting of task-relevant neuronal circuits.

Author contributions: R.R. designed research; J.V., L.L., and R.R. performed research; S.H., J.V., R.R.-P., and R.R. analyzed data; and S.H., J.V., R.R.-P., and R.R. wrote the paper.

Reviewers: F.B., Freie Universität Berlin; and T.H.D., University Medical Center Hamburg.

The authors declare no conflict of interest.

Published under the PNAS license.

¹To whom correspondence may be addressed. Email: shaegens@gmail.com or rromo@ifc.unam.mx.

This article contains supporting information online at www.pnas.org/lookup/suppl/doi:10.1073/pnas.1714633115/-DCSupplemental.

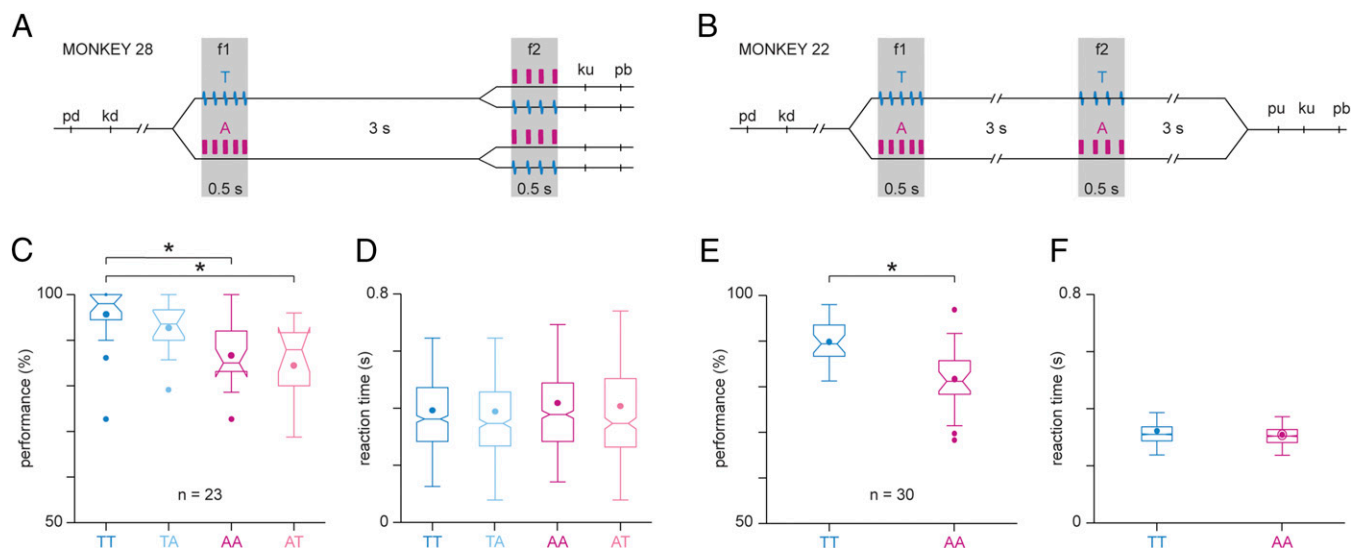


Fig. 1. Discrimination task and behavioral results. (A and B) Experimental paradigm. In m28 (A), the first and second stimuli could be either tactile (T, blue) or acoustic (A, pink), resulting in unimodal and crossmodal trials. In m22 (B), the second stimulus modality always matched with the sensory modality of the first one (unimodal trials). Conditions were randomized across trials in both monkeys. Sequence of events in the task: Mechanical probe is lowered (pd), monkey places response hand on key (kd), after a variable prestimulus delay (1–3 s) the first flutter stimulus is presented (f1, either tactile or acoustic; 500-ms duration), after a 3-s fixed delay the second stimulus is presented (f2, either tactile or acoustic, 500 ms) after which the monkey releases the key (ku), and pushes either a lateral or medial button (pb) to indicate whether f2 was of higher or lower frequency than f1, respectively. The monkey was rewarded with a drop of liquid for correct discriminations. Note that m28 was allowed to respond immediately after f2 offset (A), while m22 had to wait a 3-s fixed delay until the mechanical probe was lifted up (pu) before making the response (B). (C–F) Performance (C and E) and reaction time (D and F) box plots for each combination of stimulus modalities, for m28 (C and D) and m22 (E and F) separately. Box edges indicate 25th and 75th percentiles, central horizontal lines correspond to the median, and closed circles correspond to the mean. Vertical lines cover ± 2.7 SDs, with data points higher or lower than 2.7 SDs plotted individually (* $P < 0.05$; Kruskal–Wallis test and Bonferroni-corrected post hoc tests).

function of stimulus properties (frequency and sensory modality) and decision outcome ($f2 > f1$ or $f2 < f1$).

First, we computed the relative baseline-corrected time-frequency representations (TFRs) (SI Materials and Methods) of power, across all conditions. We observed an increase of beta-band activity during the WM delay (~ 20 - to 30 -Hz range), accompanied by an alpha- to low-beta-band decrease (~ 10 – 15 Hz), compared with baseline activity (Fig. 2). These modulations were significant in both animals (cluster- and Bonferroni-corrected $P < 0.05$). We then asked whether these oscillatory modulations were reflective of stimulus frequency and/or modality.

Stimulus Features Modulate Beta-Band Power: Frequency. To study whether the observed oscillatory activity was selective for stimulus frequency, all trials were grouped into low vs. high f1 frequency (median split per recording session). Contrasting the TFRs of these two groups, we found significantly higher beta power during the WM retention interval in trials with low vs. high stimulus frequencies (Fig. 3). This effect was present and significant (corrected $P < 0.05$) in both animals, although the time courses slightly differed: in m28, the differential beta-band response was apparent in the first half of the WM interval, whereas, in m22, beta modulation was stronger toward the second half of the WM period.

To further confirm this effect, we split trials into three conditions—low, intermediate, and high frequencies (note that trial numbers per individual frequency bin were too low to do a reliable further split)—and recomputed the beta-band time courses (Fig. S3 A and B). Indeed, there was a parametric decrease of beta power as a function of increasing f1 frequency, as revealed by regression analysis (Fig. S3 C–F; approach similar to that of ref. 19). Furthermore, we repeated the low vs. high f1 frequency split separately for both modalities, to confirm that the effect was similar in both and not driven by one modality (Fig. S3 G and H).

Stimulus Features Modulate Beta-Band Power: Modality. Next, we asked whether the observed oscillations were modulated by the stimulus modality. To address modality effects, we contrasted trials with tactile vs. acoustic stimuli, regardless of stimulus frequency. Again, we found a significant modulation of beta power in both animals (corrected $P < 0.05$), with somewhat different profiles (Fig. 4 A and C). In m28, beta-band power was significantly higher for acoustic stimulus modality compared with tactile, during both f1 and f2 periods (Fig. 4B). In m22, a more mixed pattern was observed, with beta-band power initially being stronger for the acoustic, and then for the tactile modality (Fig. 4D). Furthermore, in m22, the significant difference was maintained throughout the WM interval, whereas in m28 it was transient.

This is interesting especially in light of the fact that m22 only performed unimodal trials, in which f1 modality perfectly predicted the upcoming f2 stimulus modality, and thus provided relevant information regarding the modality to pay attention to. In contrast, m28 performed the crossmodal task, in which f1 modality did not predict f2 modality, and hence did not provide relevant (modality) information regarding the upcoming stimulus. The idea that stimulus modality is only reflected in beta power when providing relevant information, is further confirmed by the fact that, in a contrast of unimodal vs. crossmodal trials in m28, no significant differences were observed (Fig. 4 E and F).

Thus, beta-band power was reflective of relevant stimulus properties: there was a significant modulation by frequency during the WM interval, as well as a modulation by modality during stimulus presentation, which extended into the WM period only in the case of a purely unimodal task, where modality information was relevant to prepare for the upcoming second stimulus.

Beta Power Reflects Decision Outcome. Having demonstrated that beta power is reflective of stimulus features in a context-dependent manner, we then asked whether the oscillatory modulation was predictive of decision outcome. Here, we contrasted the TFRs of trials with $f2 > f1$ vs. $f2 < f1$ outcome (correct trials only) and

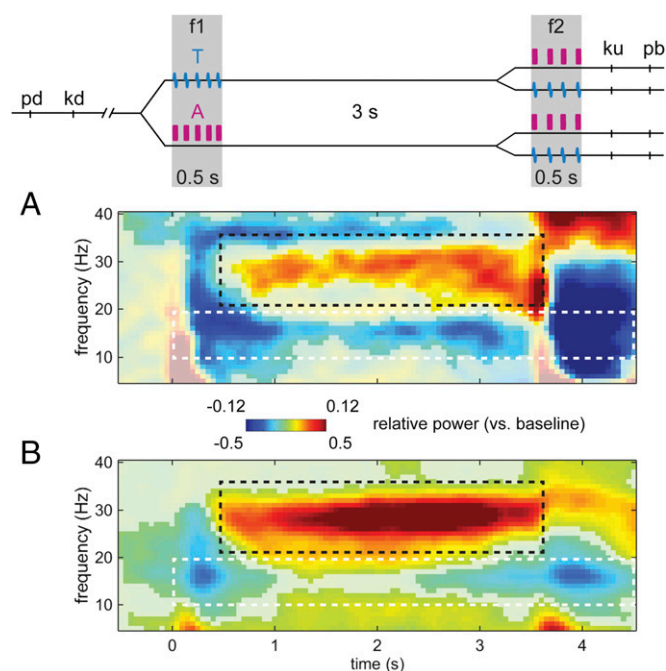


Fig. 2. Oscillatory LFP response during discrimination task. (A) Time-frequency representation of oscillatory response to the task compared with baseline activity ($t = -1$ to 0 s), averaged over all experimental trials, for m28. Time course of events as shown on Top (f1, $t = 0-0.5$ s; WM retention window, $t = 0.5-3.5$ s; f2, $t = 3.5-4$ s; response, after $t = 4$ s). Showing significant power modulations (tested vs. baseline with cluster-based permutation statistics, $P < 0.05$), averaged over all recording sessions and electrodes in MPC ($n = 130$ recording sites). Dashed boxes for reference, showing alpha/low-beta-band decrease (white) and beta increase (black). (B) Same as in A, for m22 ($n = 169$ recording sites).

found a significant modulation of beta power starting during f2 presentation (corrected $P < 0.05$). Again, we observed some differences between the two animals. In m28, beta power was higher for $f2 < f1$, and this effect only became significant after f2 offset (Fig. 5A and B), while, in m22, beta power was higher for $f2 > f1$, and the effect appeared already during f2 presentation (Fig. 5C and D).

Additionally, in m22, we separately analyzed the trials with extreme stimulus frequencies, as this animal worked with a set with wider frequency range, including very low and high f1, for which there was

no lower or higher f2 presented, respectively (cf. refs. 18 and 19). Thus, in this case, the animal could make the decision based on f1 alone, that is, effectively performing a categorization task. Taking these “categorization” trials only, we found the same beta modulation as before, now starting right after f1 offset (Fig. 5E and F). This further confirms that this particular modulation of beta power reflects the decision outcome.

Spikes Lock to the Beta Rhythm. Finally, we asked how the observed population beta oscillations interact with single-unit spikes. Taking spikes from memory-tuned cells (19), we computed spike-field coherence (SFC) (SI Materials and Methods) during the WM interval, between all simultaneously recorded spike-field pairs. We found significant SFC (corrected $P < 0.05$) matching our LFP power observations. In both animals, beta-band SFC was higher for low vs. high f1 stimulus frequencies (Fig. 6A and B), while the difference for tactile vs. acoustic was only significant in m22 (Fig. 6C and D), as we saw for power measures.

Similarly, for the decision window, we computed SFC for higher vs. lower decision outcome, and again found significant SFC modulations. Note that, for m22, we had a longer analysis window available of 3-s length, since this animal had a forced-delayed response, while for m28 we had a shorter 500-ms window, leading to a much noisier SFC estimate. Nevertheless, in both cases, we find significant modulation of SFC in the beta band for decision outcome (corrected $P < 0.05$; Fig. 6E and F). Here, both animals show the same pattern (contrary to the power modulation), with higher beta-band SFC for $f2 > f1$ than for $f2 < f1$.

Discussion

We studied the role of oscillatory activity in MPC as a function of stimulus properties and decision outcome, in two monkeys performing a tactile-acoustic discrimination task. We found that beta power as well as SFC was reflective of stimulus features in a context-dependent manner, and predictive of the decision outcome.

Timing and Context Selectivity of Beta Modulation. Beta power during the WM retention interval was reflective of f1 frequency, which had to be maintained to successfully perform the task. This finding is in line with previous studies in human subjects (13, 14, 20). The timing of the beta modulation was somewhat different between the two animals (during the first half of the WM period for m28 vs. during the last half for m22). Comparing this observation with previous results in the literature, a potential source of this difference could be the timing of the response. When the response is to be made right after f2 offset, as here for m28, the retention effect appears earlier in the WM period (13). In contrast, when there is a forced delay before the response, as here for

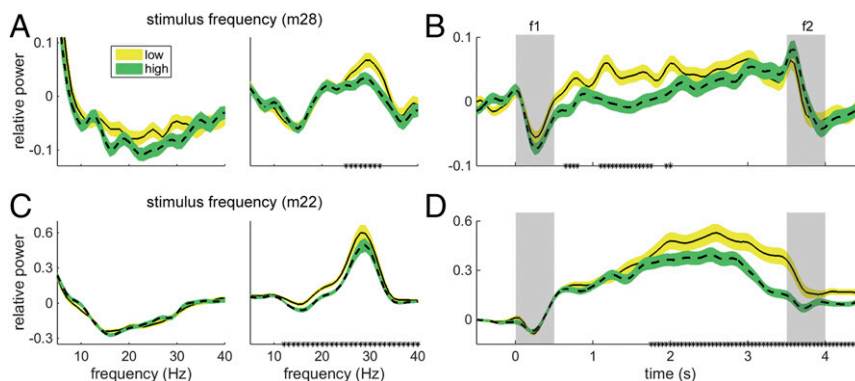


Fig. 3. Beta-band response as a function of stimulus frequency. (A) Power spectra during f1 (Left) and WM retention window (Right) showing modulation by f1 frequency, separated into low (yellow) and high (green) frequencies with median split, for m28 ($n = 130$). Significant frequency bands are marked with asterisks (cluster-based permutation test of low vs. high condition, for all time and frequency points, $P < 0.05$); colored shading reflects SEM. Note that the modulation of frequency is only significant during the WM retention window. (B) Time course of the beta-band modulation per frequency. Stimulus windows are marked by gray boxes; significant time points are marked by asterisks. (C and D) Same as A and B, for m22 ($n = 169$).

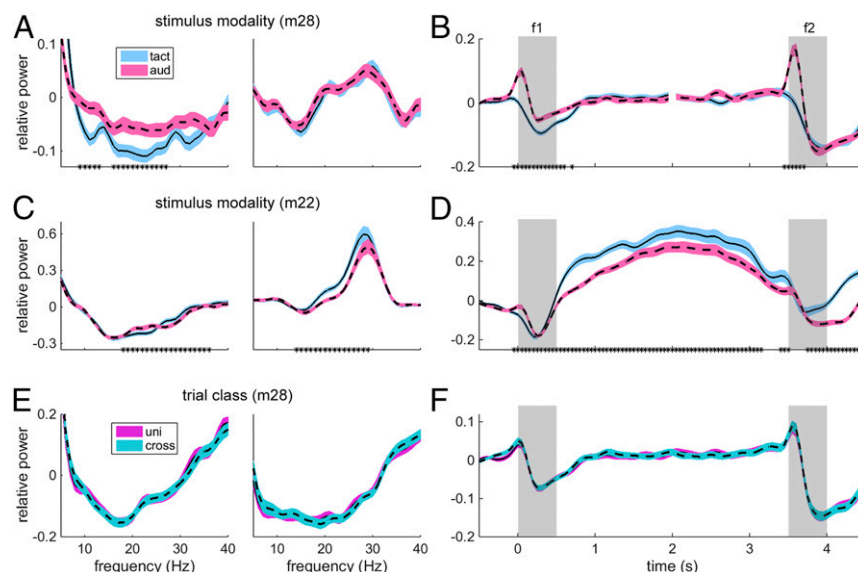


Fig. 4. Beta-band response per modality. (A) Power spectra during f1 (Left) and WM retention window (Right), for tactile (blue) and acoustic stimuli (pink), for m28 ($n = 130$). Significant frequency bands are marked with asterisks (cluster-based permutation test of tactile vs. acoustic condition, for all time and frequency points, $P < 0.05$); colored shading reflects SEM. (B) Time course of the beta-band modulation per modality. Stimulus windows are marked by gray boxes; significant time points are marked by asterisks. Note that the modulation of modality is only significant during stimulus presentation. (Note that, since m28 had both unimodal and crossmodal trials, for the time course here the first half of the plot reflects f1 modality, and the second half reflects f2 modality). (C and D) Similar to A and B, for m22 ($n = 169$). Note that, for this animal, only unimodal trials were presented, hence the difference in the beta-band time courses per modality compared with m28 (compare D vs. B). (E and F) Similar conventions as in A and B, showing modulation by trial class, separated into unimodal (purple) and crossmodal (turquoise) trials, for m28. Note that there is no significant difference between the two classes.

m22, significant modulation of beta activity is observed late in the WM delay period (15). Thus, although highly speculative at this stage, content-specific beta modulations might appear dependent on when exactly the information is task relevant (17). Alternatively, we may be picking up on different cell populations, with varying combinations of “early,” “persistent,” and “late” memory-tuned cells (19). Coincidental oversampling of one population vs.

others could then lead to slight time course differences in the grand average signals.

Perhaps in line with these putatively context-dependent temporal patterns, stimulus modality was only reflected in beta-power modulations when relevant for the task at hand. During stimulus presentation, an alpha/beta-power modulation was observed for modality, in both conditions. However, only for the unimodal

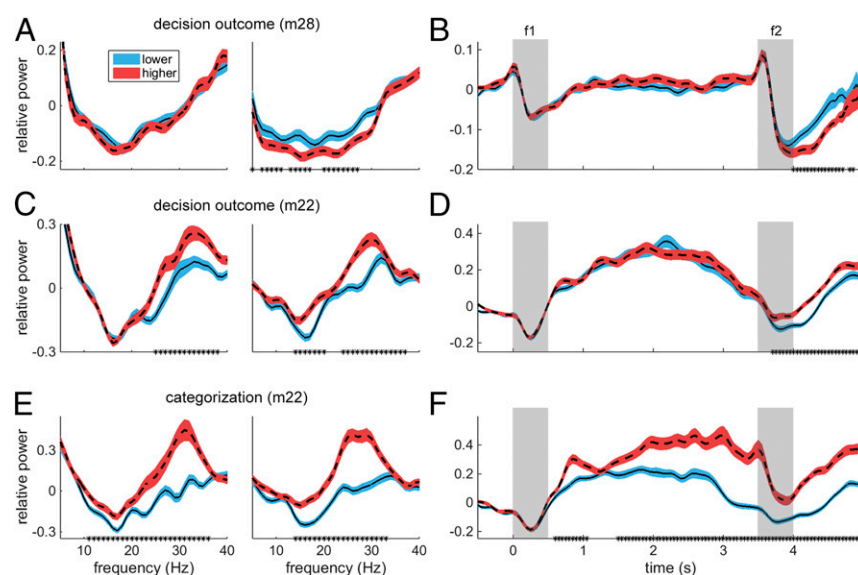


Fig. 5. Beta-band response per decision outcome. (A) Power spectra during f2 (Left) and response window (Right), for lower ($f2 < f1$, blue) and higher decision outcomes ($f2 > f1$, red), for m28 ($n = 130$). Significant frequency bands are marked with asterisks (cluster-based permutation test of lower vs. higher, correct trials only, for all time and frequency points, $P < 0.05$); colored shading reflects SEM. (B) Time course of the beta-band modulation per decision outcome. Stimulus windows are marked by gray boxes; significant time points are marked by asterisks. (C and D) Same as A and B, for m22 ($n = 169$). (E and F) Similar conventions as in A and B, showing for m22 the trials with extreme f1 frequencies (lowest vs. highest), in which the animal already knew the decision outcome after f1 presentation, since in case of lowest f1, f2 (and thus the decision outcome) could only be higher (and vice versa). Note the categorization behavior vs. the discrimination behavior observed in A–D.

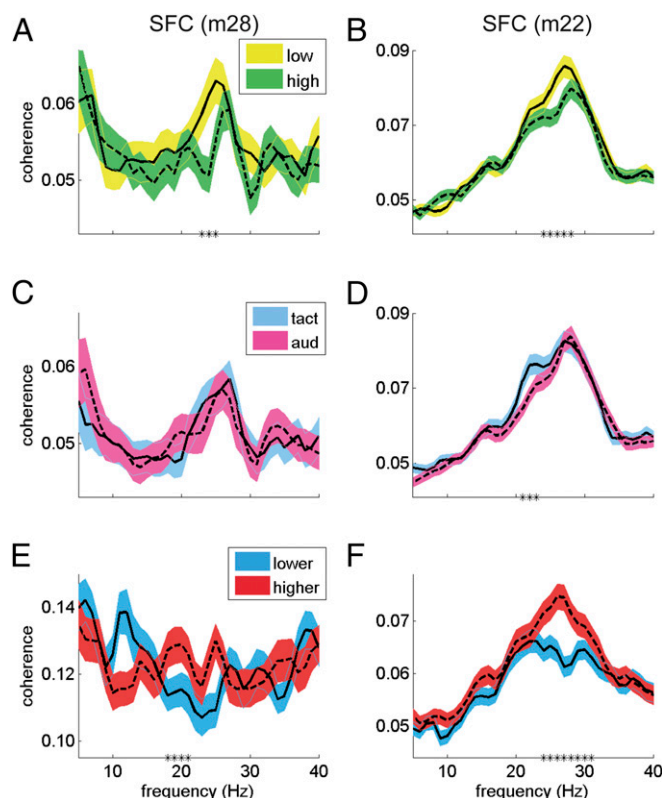


Fig. 6. Spike-field coherence per class. (A) Coherence spectra during WM retention window, for low (yellow) and high (green) stimulus frequencies, for m28 ($n = 170$ pairs). Significant frequency bands are marked with asterisks (cluster-based permutation test of low- vs. high-frequency condition, for all frequency points, $P < 0.05$); colored shading reflects SEM. (B) Same as A, for m22 ($n = 468$ pairs). (C) Similar conventions as in A, showing modulation by f1 modality, for tactile (blue) and acoustic stimuli (pink), for m28. (D) Same as C, for m22. Note that, for m22, but not for m28, there is a significant difference in beta-band SFC during WM retention for modality, similar to the effect in LFP power. (E) Similar conventions as in A, showing modulation by decision outcome during the decision window, for lower (blue) and higher decision outcomes (red), for m28. (F) Same as E, for m22.

version of the task, not for the crossmodal one, did this modulation last throughout the WM retention period. This finding suggests that information about stimulus modality is only retained when it has task relevance: in the crossmodal case, it provides no additional information since the upcoming stimulus can be in either modality, and resources are likely switched to maintaining the frequency rather than the modality of the stimulus (19); while in the unimodal case, it allows the animal to prepare specifically for the upcoming stimulus. This latter observation, while speculative, is in line with findings from a human EEG study, using a unimodal flutter discrimination paradigm with visual, tactile, and auditory stimuli, showing modality-specific modulations in the alpha band during the WM period, and a supramodal prefrontal beta signal reflecting stimulus frequency (20).

On a more cautionary note, of course we are here comparing two animals which, in addition to performing slightly different versions of the task, are simply two different individuals. Interindividual variability may also have contributed to observed differences between them.

Interpretation of Beta Modulation. Additionally, we observed a significant modulation of beta activity starting during the second stimulus presentation, which reflected the decision outcome. On a subset of categorization trials, where the decision could be made based on the first stimulus alone, the onset of this effect indeed shifted to right after the first stimulus presentation. This

finding of beta reflecting the decision outcome confirms previous reports on the same paradigm, both in monkey and human (15, 16). Furthermore, it matches findings from studies on perceptual decision making using visual or auditory paradigms (21–24).

One curious observation is that, in m28, beta power was higher for $f2 < f1$, while in m22 beta was higher for $f2 > f1$. (Although note that the SFC results did not show a sign difference between the animals.) The beta-power pattern in m22 matches a previous monkey LFP study on the tactile version of this paradigm (15), while the pattern in m28 matches a human EEG study with the same task (16). While there were some differences between the paradigms used (such as presence or absence of a forced delay before the response: by the second stimulus, m28 initiated a motor response, whereas m22 had to maintain the decision outcome for 3 more seconds), which might account for the observed differences, another parsimonious explanation could be that this sign flip simply reflects the subject's preference. Interestingly, analysis of the behavioral data indeed showed a dissociation between the animals in terms of response bias, although in opposite direction as the beta modulation: m28 had a slight bias for answering “higher,” while m22 had a bias for “lower.” Whether and how these observations are connected remains to be seen, but one potential explanation might be phasic neuromodulation, which has been linked on the one hand to suppression of decision bias (25, 26), and on the other hand to beta oscillatory activity (e.g., refs. 27 and 28).

Alternatively, the beta signal might reflect the summed activation of different (motor) networks; depending on which response is selected, and due to uneven sampling across these networks, one dominates the signal over the other (17, 29). In either case, while the sign of the effect may not be meaningful itself, the relative value is, as it predicts the subject's decision (15).

Which brings us to the key question: what mechanistic role does beta play here? When we report that beta-power modulations reflect the stimulus properties or decision outcome, we do not mean to suggest that beta per se encodes this information. Many reports show convincingly that the actual information is encoded by spike firing rates and other spike codes (2, 7, 8, 10, 19, 30–32). What we might be picking up on here are the local neuronal ensembles—encompassing the single units in which we observe (for example) rate coding—combining their activity into a population code. Fluctuations in the LFP then reflect shifts in these activated local networks. Our SFC findings support such an account: single-unit spike activity of cells that maintain stimulus frequency during the WM delay [as revealed by regression analysis (19)] was phase-locked to the beta cycle, and this beta-band coherence was selectively modulated as a function of relevant task features.

Recently, Spitzer and Haegens (17) proposed that beta synchronization provides a mechanism for the flexible, transient formation of functional neuronal ensembles. In this framework, a modulation in beta power reflects the endogenous (re)activation of a task-relevant cortical representation, in the time window when information contained in this circuit is required for subsequent processing steps. This view is in line with the idea that beta facilitates (long-range) communication between networks (33–36), and specifically top-down-driven interactions (37–40). Furthermore, recent computational modeling work (41, 42) suggests that cortical (somatomotor) beta oscillations are generated via the integration of concurrent inputs along the proximal and distal locations of the apical dendrites of pyramidal cells (including both feedforward input via the granular layer and feedback drives via supragranular layers), thus providing a potential mechanistic implementation for the top-down-driven synchronization of task-relevant cell assemblies.

Our current findings, especially in light of the previously reported spike results from these same recordings (19), fit this proposed framework (17) well. While large-scale concurrent spike and LFP recordings are required to further test these proposed mechanisms, our findings provide tentative evidence for a role for beta in ensemble formation: (i) we find local population beta-power modulations corresponding to task-relevant WM content and decision outcome, (ii) complementing previous findings based on spike recordings in this same study, and (iii) these patterns are

directly linked as shown by our SFC results (i.e., memory-tuned spikes are coupled to beta oscillations). Furthermore, as alluded to above, when animals performed versions of the task with different timing (immediate vs. delayed response), the temporal pattern of the beta modulation shifted accordingly. This might reflect the time-dependent recruitment of relevant frontal lobe circuits based on task demands.

Conclusion

To summarize, here, we report content-specific beta-power modulations in the premotor system, reflecting task-relevant stimulus features maintained in WM, and subsequent decision outcomes. This information is coded in a supramodal, context-dependent manner—modality information is only retained when relevant for the task at hand. We propose that these beta modulations are a signature of the formation of functional neuronal ensembles, which encode task-relevant information in (population) spike activity. These findings confirm previous work on somatosensory perceptual decision making (13–16, 43) and extend them to the supramodal case (20); that is, the observed modulations are not exclusive to somatosensory processing. Furthermore, our results nicely fit a recently proposed framework in which content-specific beta synchronization provides a mechanism for the formation of functional neuronal ensembles during endogenous (re)activation of

task-relevant cortical representations (17). Future work, including large-scale recordings with high spatial resolution, should test specific aspects of this framework, such as its spatial extent and temporal control.

Materials and Methods

Two monkeys (*Macaca mulatta*), referred to as m28 and m22, were trained to perform a discrimination task (1, 18, 19), in which they had to discriminate the difference in frequency between two sequentially delivered flutter stimuli (Fig. 1 A and B). Both spikes and LFPs were recorded simultaneously from the MPC (Fig. S1). Details for data acquisition and analysis are provided in *SI Materials and Methods*. Animals were handled in accordance with the standards of the National Institutes of Health and the Society for Neuroscience. All protocols were approved by the Institutional Animal Care and Use Committee of the Instituto de Fisiología Celular of the National Autonomous University of México (UNAM).

ACKNOWLEDGMENTS. This work was supported by Netherlands Organization for Scientific Research Veni Grant 451-14-027 (to S.H.), Dirección de Asuntos del Personal Académico de la UNAM, Consejo Nacional de Ciencia y Tecnología, and Fondo Jaime Torres Bodet de la Secretaría de Educación Pública, México (R.R.). J.V. is a doctoral student from Programa de Doctorado en Ciencias Biomédicas, UNAM, and received Fellowship 255865 from Consejo Nacional de Ciencia y Tecnología.

- Hernández A, Salinas E, García R, Romo R (1997) Discrimination in the sense of flutter: New psychophysical measurements in monkeys. *J Neurosci* 17:6391–6400.
- Romo R, Salinas E (2003) Flutter discrimination: Neural codes, perception, memory and decision making. *Nat Rev Neurosci* 4:203–218.
- Romo R, Lemus L, de Lafuente V (2012) Sense, memory, and decision-making in the somatosensory cortical network. *Curr Opin Neurobiol* 22:914–919.
- Salinas E, Hernández A, Zainos A, Romo R (2000) Periodicity and firing rate as candidate neural codes for the frequency of vibrotactile stimuli. *J Neurosci* 20:5503–5515.
- Luna R, Hernández A, Brody CD, Romo R (2005) Neural codes for perceptual discrimination in primary somatosensory cortex. *Nat Neurosci* 8:1210–1219.
- Brody CD, Hernández A, Zainos A, Romo R (2003) Timing and neural encoding of somatosensory parametric working memory in macaque prefrontal cortex. *Cereb Cortex* 13:1196–1207.
- Barak O, Tsodyks M, Romo R (2010) Neuronal population coding of parametric working memory. *J Neurosci* 30:9424–9430.
- Romo R, Hernández A, Zainos A, Lemus L, Brody CD (2002) Neuronal correlates of decision-making in secondary somatosensory cortex. *Nat Neurosci* 5:1217–1225.
- Lemus L, et al. (2007) Neural correlates of a postponed decision report. *Proc Natl Acad Sci USA* 104:17174–17179.
- Hernández A, et al. (2010) Decoding a perceptual decision process across cortex. *Neuron* 66:300–314.
- Haegens S, Osipova D, Oostenveld R, Jensen O (2010) Somatosensory working memory performance in humans depends on both engagement and disengagement of regions in a distributed network. *Hum Brain Mapp* 31:26–35.
- Haegens S, Nacher V, Luna R, Romo R, Jensen O (2011) α -Oscillations in the monkey sensorimotor network influence discrimination performance by rhythmic inhibition of neuronal spiking. *Proc Natl Acad Sci USA* 108:19377–19382.
- Spitzer B, Wacker E, Blankenburg F (2010) Oscillatory correlates of vibrotactile frequency processing in human working memory. *J Neurosci* 30:4496–4502.
- Spitzer B, Blankenburg F (2011) Stimulus-dependent EEG activity reflects internal updating of tactile working memory in humans. *Proc Natl Acad Sci USA* 108:8444–8449.
- Haegens S, et al. (2011) Beta oscillations in the monkey sensorimotor network reflect somatosensory decision making. *Proc Natl Acad Sci USA* 108:10708–10713.
- Herding J, Spitzer B, Blankenburg F (2016) Upper beta band oscillations in human premotor cortex encode subjective choices in a vibrotactile comparison task. *J Cogn Neurosci* 28:668–679.
- Spitzer B, Haegens S (2017) Beyond the status quo: A role for beta oscillations in endogenous content (re)activation. *eNeuro* 0170-17.2017.
- Lemus L, Hernández A, Luna R, Zainos A, Romo R (2010) Do sensory cortices process more than one sensory modality during perceptual judgments? *Neuron* 67:335–348.
- Vergara J, Rivera N, Rossi-Pool R, Romo R (2016) A neural parametric code for storing information of more than one sensory modality in working memory. *Neuron* 89:54–62.
- Spitzer B, Blankenburg F (2012) Supramodal parametric working memory processing in humans. *J Neurosci* 32:3287–3295.
- Donner TH, Siegel M, Fries P, Engel AK (2009) Buildup of choice-predictive activity in human motor cortex during perceptual decision making. *Curr Biol* 19:1581–1585.
- Gould IC, Nobre AC, Wyart V, Rushworth MFS (2012) Effects of decision variables and intraparietal stimulation on sensorimotor oscillatory activity in the human brain. *J Neurosci* 32:13805–13818.
- Wyart V, de Gardelle V, Scholl J, Summerfield C (2012) Rhythmic fluctuations in evidence accumulation during decision making in the human brain. *Neuron* 76:847–858.
- Kubaneck J, Snyder LH, Brunton BW, Brody CD, Schalk G (2013) A low-frequency oscillatory neural signal in humans encodes a developing decision variable. *Neuroimage* 83:795–808.
- de Gee JW, Knapen T, Donner TH (2014) Decision-related pupil dilation reflects upcoming choice and individual bias. *Proc Natl Acad Sci USA* 111:E618–E625.
- de Gee JW, et al. (2017) Dynamic modulation of decision biases by brainstem arousal systems. *eLife* 6:e23232.
- Belitski A, et al. (2008) Low-frequency local field potentials and spikes in primary visual cortex convey independent visual information. *J Neurosci* 28:5696–5709.
- Donner TH, Siegel M (2011) A framework for local cortical oscillation patterns. *Trends Cogn Sci* 15:191–199.
- Wimmer K, Ramon M, Pasternak T, Compté A (2016) Transitions between multiband oscillatory patterns characterize memory-guided perceptual decisions in prefrontal circuits. *J Neurosci* 36:489–505.
- Romo R, Brody CD, Hernández A, Lemus L (1999) Neuronal correlates of parametric working memory in the prefrontal cortex. *Nature* 399:470–473.
- Hernández A, Zainos A, Romo R (2002) Temporal evolution of a decision-making process in medial premotor cortex. *Neuron* 33:959–972.
- Romo R, de Lafuente V (2013) Conversion of sensory signals into perceptual decisions. *Prog Neurobiol* 103:41–75.
- Kopell N, Ermentrout GB, Whittington MA, Traub RD (2000) Gamma rhythms and beta rhythms have different synchronization properties. *Proc Natl Acad Sci USA* 97:1867–1872.
- Varela F, Lachaux J-P, Rodriguez E, Martinerie J (2001) The brainweb: Phase synchronization and large-scale integration. *Nat Rev Neurosci* 2:229–239.
- Benchenane K, Tiesinga PH, Battaglia FP (2011) Oscillations in the prefrontal cortex: A gateway to memory and attention. *Curr Opin Neurobiol* 21:475–485.
- Siegel M, Engel AK, Donner TH (2011) Cortical network dynamics of perceptual decision-making in the human brain. *Front Hum Neurosci* 5:21.
- Engel AK, Fries P (2010) Beta-band oscillations—signalling the status quo? *Curr Opin Neurobiol* 20:156–165.
- Wang X-J (2010) Neurophysiological and computational principles of cortical rhythms in cognition. *Physiol Rev* 90:1195–1268.
- Arnal LH, Giraud A-L (2012) Cortical oscillations and sensory predictions. *Trends Cogn Sci* 16:390–398.
- Bastos AM, et al. (2012) Canonical microcircuits for predictive coding. *Neuron* 76:695–711.
- Jones SR, et al. (2009) Quantitative analysis and biophysically realistic neural modeling of the MEG mu rhythm: Rhythmogenesis and modulation of sensory-evoked responses. *J Neurophysiol* 102:3554–3572.
- Sherman MA, et al. (2016) Neural mechanisms of transient neocortical beta rhythms: Converging evidence from humans, computational modeling, monkeys, and mice. *Proc Natl Acad Sci USA* 113:E4885–E4894.
- Herding J, Ludwig S, Blankenburg F (2017) Response-modality-specific encoding of human choices in upper beta-band oscillations during vibrotactile comparisons. *Front Hum Neurosci* 11:118.
- Hernández A, et al. (2008) Procedure for recording the simultaneous activity of single neurons distributed across cortical areas during sensory discrimination. *Proc Natl Acad Sci USA* 105:16785–16790.
- Oostenveld R, Fries P, Maris E, Schoffelen J-M (2011) FieldTrip: Open source software for advanced analysis of MEG, EEG, and invasive electrophysiological data. *Comput Intell Neurosci* 2011:156869.
- Fries P, Womelsdorf T, Oostenveld R, Desimone R (2008) The effects of visual stimulation and selective visual attention on rhythmic neuronal synchronization in macaque area V4. *J Neurosci* 28:4823–4835.
- Maris E, Oostenveld R (2007) Nonparametric statistical testing of EEG- and MEG-data. *J Neurosci Methods* 164:177–190.